Stable Higher Categories

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The goal of todays talk is to introduce stable higher categories. We first start by reminding ourselves of its historical predecessor: triangulated categories. For the rest we focus on stable higher categories and various equivalent ways it can be defined. Most of the material here is taken from [Gr12].

Some History: Triangulated Categories

Triangulated Categories were defined during Verdier's 1963 thesis under Grothendieck. They were developed in order to describe the structure present in the derived category of an abelian category.

Definition 1.1. A *triangulated category* consists of

- an additive category $\mathcal{D}$;
- an equivalence of categories $T : \mathcal{D} \to \mathcal{D}$ called a *translation functor*, where we write $T^k(d) = d[k]$; and
- a collection of *distinshuished triangles*

$$X \xrightarrow{f} Y \xrightarrow{g} Z \xrightarrow{h} X[1].$$

We require these to satisfy the following axioms.

TR1 (a) Every morphism in $\mathcal{D}$ is part of a distinguished triangle.

(b) The collection of distinquished triangles is stable under isomorphism.

(c) For every object X in $\mathcal{D}$, we have a distinguished triangle

$$X \xrightarrow{1_X} X \longrightarrow 0 \longrightarrow X[1].$$

TR2 The triangle

$$X \xrightarrow{f} Y \xrightarrow{g} Z \xrightarrow{h} X[1]$$

is distinguished if and only if the *rotated* triangle

$$Y \xrightarrow{g} Z \xrightarrow{h} X[1] \xrightarrow{-f[1]} Y[1]$$

is.

TR3 Given the solid commutative diagram

$$\begin{tikzcd}
X \arrow[r] \arrow[d, "\phi"'] & Y \arrow[r] \arrow[d] & Z \arrow[r] \arrow[d, dotted] & X[1] \arrow[d, "\phi[1]"] \\
X' \arrow[r] & Y' \arrow[r] & Z' \arrow[r] & X'[1]
\end{tikzcd}$$

where the rows are distinguished triangles, there exists a morphism $Z \to Z'$ completing the diagram.

TR4 (Octahedral Axiom) Let

$$X \xrightarrow{f} Y \xrightarrow{u} Y/X \xrightarrow{d} X[1]$$

$$Y \xrightarrow{g} Z \xrightarrow{v} Z/Y \xrightarrow{d'} Y[1]$$

$$X \xrightarrow{gf} Z \xrightarrow{w} Z/X \xrightarrow{d''} X[1]$$

be distinguished triangles. Then there is a fourth distinguished triangle

$$Y/X \xrightarrow{\phi} Z/X \xrightarrow{\psi} Z/Y \xrightarrow{\theta} Y/X[1]$$

such that the diagram

$$\begin{tikzcd}
X \arrow[rr, "gf"] \arrow[dr, "f"] & & Z \arrow[rr, "v"] \arrow[dr, "w"] & & Z/Y \arrow[rr, "\theta"] \arrow[dr, "d'"] & & Y/X[1] \\
& Y \arrow[ur, "g"] \arrow[dr, "u"] & & Z/X \arrow[ur, "\psi"] \arrow[dr, "d''"] & & Y[1] \arrow[ur, "u[1]"] & \\
& & Y/X \arrow[ur, "\phi"] \arrow[rr, "d"] & & X[1] \arrow[ur, "f[1]"] & &
\end{tikzcd}$$

is commutative.

One of the main examples of a triangulated category is the derived category of an abelian category. The translation functor is given by shifting chain complexes and chain maps. The distinguished triangles are exactly the exact triangles.

Another example of a triangulated category is the homotopy category of spectra. Translation is given by suspension, while the distinguished triangles are the fibre sequences (which are the same thing as cofibre sequences).

The notion of a triangulation on a category proved to be useful enough for people to keep using it, but it had some serious drawbracks. Perhaps the most

fundamental drawback is the fact that the structure of a triangulation requires one to specify a huge amount of data, very little of which is functorial.

It turns out that most triangulated categories that appear in practice come as the homotopy category of a stable ∞-category. (However, one important example of a triangulated category not arising in this way is the derived category of mixed motives.) It is perhaps more natural to work with this stable ∞-category instead.

First Definition of a Stable Higher Category

As it almost always seems to be the case, we first need a flurry of definitions before we can give our desired definition.

Definition 2.1. An object $0 \in \mathcal{D}$ is called a *zero object* if it is both initial and final. A quasi-category is called *pointed* if it has a zero object.

Definition 2.2. A *triangle* in $\mathcal{D}$ is a diagram

$$\begin{tikzcd} X \arrow[r, "f"] \arrow[d] & Y \arrow[d, "g"] \\ 0 \arrow[r] & Z \end{tikzcd}$$

where 0 is a zero object of $\mathcal{D}$. We will call such a triangle a *fiber sequence* (resp. *cofiber sequence*) if it is a pullback diagram (resp. pushout diagram), and in this case we will say that X is the *fibre* of g (resp. Z is the *cofibre* of f).

Definition 2.3. An ∞-category $\mathcal{D}$ is *stable* if

ST1 it is pointed

ST2 every morphism in $\mathcal{D}$ admits a fibre and a cofibre.

ST3 a triangle in $\mathcal{D}$ is a fibre sequence if and only if it is a cofibre sequence.

Remark 2.4. Immediately, one should note that stablity is a *property* of an ∞-category, not a piece of additional structure. This suggests that stable ∞-categories are somewhat more natural than triangulated categories.

One of the beauties of this definition is that we can now use the power of quasi-categories to deduce functoriality of our construction

Functoriality of Cofibres

As a first consequence of the definition, let's take a look at one of the advertised advantages of stable ∞-categories: functorialty of cofibres. To begin, we should state a simplified version of a proposition of Lurie's *Higher Topos Theory.* [4.3.2.15]

Proposition 3.1. *Let $\mathcal{C}$ and $\mathcal{D}$ be ∞-categories, and let $\mathcal{C}^0$ be a full subcategory of $\mathcal{C}$. Let $\mathcal{K} \subseteq \mathrm{Fun}(\mathcal{C}, \mathcal{D})$ be the full subcategory spanned by functors $F : \mathcal{C} \to \mathcal{D}$ which are left Kan extensions of $F|_{\mathcal{C}^0}$. Let $\mathcal{K}' \subseteq \mathrm{Fun}(\mathcal{C}^0, \mathcal{D})$ be the full subcategory spanned by functors $F_0 : \mathcal{C}^0 \to \mathcal{D}$ with the property that, for each object C in $\mathcal{C}$, the induced diagram $C_{/C} \times_{\mathcal{C}} \mathcal{C}^0 \to \mathcal{D}$ has a colimit. Then the restriction functor $\mathcal{K} \to \mathcal{K}'$ is a trivial fibration.*

For notational convenience, we will write $C^0_{/C}$ for $C_{/C} \times_{\mathcal{C}} \mathcal{C}^0$. We haven't talked about Kan extensions in the context of ∞-category theory, but let us take for granted that there is some suitable generalization. The moral content of this proposition is that left Kan extensions can be computed using certain colimits and that they are unique up to a contractible space of choices.

We will now apply this proposition twice in quick succession to construct functorial cofibre functor $\mathrm{cofib} : \mathrm{fun}(\Delta^1, \mathcal{D}) \to \mathcal{D}$ for a stable ∞-category $\mathcal{D}$. For the first application, take

$$\begin{aligned} \mathcal{C} &= \Delta^1 \vee \Delta^1 \\ \mathcal{C}^0 &= \Delta^1. \end{aligned}$$

In this case, we get that $\mathcal{K} \subseteq \mathrm{Fun}(\mathcal{C}, \mathcal{D})$ is the full subcategory spanned by diagrams of the form

$$\begin{array}{ccc} \bullet & \longrightarrow & \bullet \\ \downarrow & & \\ 0 & & \end{array}$$

and $\mathcal{K}' \subseteq \mathrm{Fun}(\mathcal{C}^0, \mathcal{D})$ is the full category, *i.e.* diagrams of the form

$$\bullet \longrightarrow \bullet$$

and the proposition tells us that the restriction functor

$$\left\{ \begin{array}{ccc} \bullet & \longrightarrow & \bullet \\ \downarrow & & \\ 0 & & \end{array} \right\} \to \{ \bullet \longrightarrow \bullet \}$$

is a trivial fibration.

For the second applicaiton, take

$$\begin{aligned} \mathcal{C} &= \Delta^1 \times \Delta^1 \\ \mathcal{C}^0 &= \Delta^1 \vee \Delta^1. \end{aligned}$$

In this case, we have that $\mathcal{K} \subseteq \mathrm{Fun}(\mathcal{C}, \mathcal{D})$ is the full subcategory spanned by pushout squares

$$\begin{array}{ccc} \bullet & \longrightarrow & \bullet \\ \downarrow & \ulcorner & \downarrow \\ \bullet & \longrightarrow & \bullet \end{array}$$

and $\mathcal{K}' \subseteq \mathrm{Fun}(\mathcal{C}^0, \mathcal{D})$ is the full category spanned by diagrams that can be completed to a pushout square

$$\begin{array}{ccc} \bullet & \longrightarrow & \bullet \\ \downarrow & \ulcorner & \downarrow \\ \bullet & \dashrightarrow & \bullet \end{array}$$

and the proposition tells us that the restriction functor

$$\left\{\begin{tikzcd} \bullet \arrow[r] \arrow[d] & \bullet \arrow[d] \\ \bullet \arrow[r] \arrow[ur, phantom, "\ulcorner"] & \bullet \end{tikzcd}\right\} \to \left\{\begin{tikzcd} \bullet \arrow[r] \arrow[d] & \bullet \arrow[d, dotted] \\ \bullet \arrow[r, dotted] \arrow[ur, phantom, "\ulcorner"] & \bullet \end{tikzcd}\right\}$$

is a trivial fibration.

Now, we can pullback this last trivial fibration along the fully faithful functor

$$\left\{\begin{tikzcd} \bullet \arrow[r] \arrow[d] & \bullet \arrow[d, dotted] \\ 0 \arrow[r, dotted] \arrow[ur, phantom, "\ulcorner"] & \bullet \end{tikzcd}\right\} \to \left\{\begin{tikzcd} \bullet \arrow[r] \arrow[d] & \bullet \arrow[d, dotted] \\ \bullet \arrow[r, dotted] \arrow[ur, phantom, "\ulcorner"] & \bullet \end{tikzcd}\right\}$$

to get a trivial fibration

$$\left\{\begin{tikzcd} \bullet \arrow[r] \arrow[d] & \bullet \arrow[d] \\ 0 \arrow[r] \arrow[ur, phantom, "\ulcorner"] & \bullet \end{tikzcd}\right\} \to \left\{\begin{tikzcd} \bullet \arrow[r] \arrow[d] & \bullet \arrow[d, dotted] \\ 0 \arrow[r, dotted] \arrow[ur, phantom, "\ulcorner"] & \bullet \end{tikzcd}\right\}.$$

Collecting all the relevant morphisms, we get chain of functors

$$\left\{\begin{tikzcd} \bullet \arrow[r] \arrow[d] & \bullet \arrow[d] \\ 0 \arrow[r] \arrow[ur, phantom, "\ulcorner"] & \bullet \end{tikzcd}\right\} \to \left\{\begin{tikzcd} \bullet \arrow[r] \arrow[d] & \bullet \arrow[d, dotted] \\ 0 \arrow[r, dotted] \arrow[ur, phantom, "\ulcorner"] & \bullet \end{tikzcd}\right\} \to \left\{\begin{tikzcd} \bullet \arrow[r] \arrow[d] & \bullet \\ 0 & \end{tikzcd}\right\} \to \{ \bullet \to \bullet \}$$

where the first and third maps are trivial fibrations and the second is the inclusion of a full subcategory. However, since $\mathcal{D}$ is stable and every morphism admits a cofibre, the second map is in fact the identity functor, so the composition is a trivial fibration. We can now compose the restriction to the final vertex with a section of this trivial fibration to obtain the desired functorial cofibre functor. Fibre can be chosen functorially in exactly the same manner.

Loop and Suspension Functors

Loops and suspensions where they were originally defined as some special push outs along cones (or pullbacks along paths). In the realm of higher categories, however, no such care is necessary as the notion of a colimits and limits in higher categories already takes care of all those issues. Thus we have following definitions:

Definition 4.1. Let $\mathcal{C}$ be a pointed quasi-category with pullbacks and X an object. Then *loops on* X, denoted by ΩX, is defined by the pullback

$$\begin{tikzcd} \Omega X \arrow[r] \arrow[d] \arrow[dr, phantom, "\ulcorner"] & * \arrow[d] \\ * \arrow[r] & X \end{tikzcd}$$

Definition 4.2. Let $\mathcal{C}$ be a pointed quasi-category with pushouts and X an object. Then *suspension of* X, denoted by ΣX, is defined by the pushout

$$\begin{tikzcd} X \arrow[r] \arrow[d] \arrow[dr, phantom, "\ulcorner", very near start] & * \arrow[d] \\ * \arrow[r] & \Sigma X \end{tikzcd}$$

As a stable quasi-category is all fibres and cofibres it certainly has all loops and suspensions. In fact using the fact that cofibre and fibre sequences are equivalent we immediately get following result:

Proposition 4.3. *Let $\mathcal{D}$ be a stable quasi-category. Then the natural maps*

$$X \to \Omega\Sigma X$$

$$\Sigma\Omega X \to X$$

are equivalences.

In fact we can make this construction into a whole functors

$$\Omega : \mathcal{D} \to \mathcal{D} \quad \Sigma : \mathcal{D} \to \mathcal{D}$$

The loop and suspension functors on a stable ∞-category $\mathcal{D}$ are defined again using Proposition 3.1. We use it to construct a functor $\mathcal{D} \to \mathrm{Fun}(\Delta^1, \mathcal{D})$ that takes every object X to a morphism $X \to 0$. We then compose this with the cofibre functor to get the suspension functor Σ.

We can construct a loop functor Ω in exactly the same way. Having built two functors we realize that they actually form an adjoint pair:

$$\begin{tikzcd} \mathcal{D} \arrow[r, shift left, "\Sigma"] & \mathcal{D} \arrow[l, shift left, "\Omega"] \end{tikzcd}$$

Moreover, since $\mathcal{D}$ is stable, from what we showed above we get that Ω and Σ form an adjoint equivalence. This condition is in fact sufficient as we have following result:

Proposition 4.4. *Let $\mathcal{D}$ be a pointed quasi-category. Then $\mathcal{D}$ is stable if and only if the adjunction*

$$\begin{tikzcd} \mathcal{D} \arrow[r, shift left, "\Sigma"] & \mathcal{D} \arrow[l, shift left, "\Omega"] \end{tikzcd}$$

is an actual equivalence.

Notation 4.5. We will often write $[n] : \mathcal{D} \to \mathcal{D}$ to denote the nth power of suspension.

Alternative Characterizations of Stability

Given two objects X, Y in a stable infinite category $\mathcal{D}$, we can define their coproduct as

$$X \sqcup Y = \mathrm{cofib}(0 : X[-1] \to Y).$$

Given two morphisms $f, g : X \to Y$, we can compute their coequalizer as

$$\mathrm{coeq}(f, g) = \mathrm{cofib}(f - g)$$

where we have used $f - g$ to denote a morphism that represents the difference in the homotopy category. We thus have that $\mathcal{D}$ admits finite colimits. And analogous argument can be used to show that $\mathcal{D}$ admits finite limits. This proves the first part of the following proposition. The second follows from two applications of Proposition 3.1.

Proposition 5.1. *A pointed ∞-category $\mathcal{D}$ is stable if and only if*

1. *$\mathcal{D}$ admits finite limits and colimits, and*
2. *a square*

$$\begin{tikzcd} X \arrow[r] \arrow[d] & Y \arrow[d] \\ Z \arrow[r] & W \end{tikzcd}$$

in $\mathcal{D}$ is a pushout if and only if it is a pullback.

Perhaps surprisingly, we also have the following characterization, suggesting that stability is some sort of generalization of pointedness

Proposition 5.2. *An ∞-category is stable if and only if*

1. *it admits finite limits and colimits, and*
2. *finite limits and colimits commute.*

Proof. We will only prove the backwards implication. Suppose

$$\begin{tikzcd} A \arrow[r] \arrow[d] & X \arrow[d] \\ B \arrow[r] & Y \end{tikzcd}$$

is a pushout diagram. Then consider the following diagram.

$$\begin{tikzcd} A \arrow[d] & A \arrow[l] \arrow[r] \arrow[d] & B \arrow[d] \\ X & A \arrow[l] \arrow[r] & B \\ X \arrow[u] & A \arrow[l] \arrow[r] \arrow[u] & A \arrow[u] \end{tikzcd}$$

If we compute the pullbacks of the columns and then the pushout of the resulting diagram, we get A. If instead we compute the pushout of the rows and then the pullback of the result, we get $B \times_Y X$. Thus, the pushout square is also a pullback square. The rest of the proof procedes similarly. $\square$

Remark 5.3. Notice how interesting this condition is. In every other definition of stable quasi-categories we proceeded in two steps. First we defined pointed quasi-categories and then we used the fact that it is pointed to add the conditions we desired. The last condition, however, has no two parts. It is defined directly.

References

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